

A polynomial-dimensional refinement of sharp simplex stability

AI-assisted research supplement to entry005

7 October 2026

Abstract

For the projection-cone invariant of the pinned source manuscripts, the sharp exponent $1/(d-1)$ in the centroid containment estimate for every maximum-volume inscribed simplex admits an explicit coefficient bounded by $2^{20}d^6$. The proof uses first absolute determinant moments, a weighted selection of support anchors, the actual convex-body cone law, projection caps, and retention of the prescribed maximum simplex. An explicit square-pyramid construction gives the necessary coefficient bound $(d+1)[d(d+1)]^{1/(d-1)}$. This is a typeset derivative of the frozen written refinement and its companion lower-bound proof; it makes no first-discovery, best-known-result, human peer-review, or proof-assistant formalization claim.

Result and conventions

A convex body is a compact convex set with nonempty interior. We use exactly the invariant and excess of the pinned manuscripts. The projection body ΠK is defined by

$$h_{\Pi K}(u) = |K \mid u^\perp|_{d-1} \quad (\|u\| = 1),$$

where the right-hand side is the $(d-1)$ -dimensional volume of the orthogonal projection. The operator $\mathcal{P}$ denotes a pyramid over K , with the convention in entry005 v3, Section 1 [2]. In particular, $\mathcal{P}K$ has dimension $d+1$; the projection ratio is evaluated in the dimension of its argument. These are the source definitions, not a different projection-body invariant. Put

$$\begin{aligned} R_{\text{proj}}(K) &= \frac{|\Pi K|}{|K|^{d-1}}, \\ a(K) &= \left(\frac{d}{d+1}\right)^d \frac{R_{\text{proj}}(\mathcal{P}K)}{R_{\text{proj}}(K)} - 1, \\ e(K) &= a(K) - \frac{1}{d+1}, \\ E(K, S) &= \inf\{t \geq 0 : K \subset z_S + (1+t)(S - z_S)\}, \end{aligned}$$

where z_S is the centroid of S .

Theorem. *For every $d \geq 3$, every full-dimensional convex body $K \subset \mathbb{R}^d$, and every maximum-volume inscribed simplex S ,*

$$E(K, S) \leq G_d e(K)^{1/(d-1)} \leq 2^{20} d^6 e(K)^{1/(d-1)}.$$

The exponent is unchanged and optimal, by the actual simplex-truncation family in the pinned manuscript [1, 4].

The exact constants are

$$\begin{aligned}
m &= d - 1, & R &= d\sqrt{d+2}, & M &= 4dR, \\
C &= (d+1)^2(d+2), & L &= 2\sqrt{m}(R+1), \\
J &= \frac{d}{2}(2R)^d[1 + M + d2^{d-1}M], \\
G_d &= 8(R+1)dL(JC)^{1/m}, & e_0 &= \frac{1}{JC(8dL)^m}.
\end{aligned}$$

For $0 \leq e \leq e_0$, the stronger local estimate with coefficient $G_d/2$ holds. Moreover,

$$G_d e_0^{1/m} = R + 1,$$

which makes the global branch transparent.

The main structural strengthening is valid for probability laws without any convex-body realization assumption: a centered full-dimensional probability law of bounded support admits an assignment to $d+1$ affinely independent support points with mean distance at most

$$\text{diam}(\text{supp } \nu) \frac{(d+1)(d+2)}{2} \frac{B-A}{B}.$$

Distances and diameter may be taken in any fixed norm. This assertion requires no dispersion assumption, inverse determinant, covariance estimate, second determinant moment, or small-probability conditioning loss. The refinement is relative to the supplied argument; no first-discovery or best-known-result claim is made.

1 The stronger integrated assignment theorem

Let ν be centered with bounded support spanning $\mathbb{R}^d$. Let $X_1, X_2, \dots$ be independent with law ν , and define the *first absolute determinant moments*

$$\begin{aligned}
A &= \mathbb{E} |\det(X_1, \dots, X_d)|, \\
B &= \mathbb{E} |\det((X_0, 1), \dots, (X_d, 1))|, & D &= B - A.
\end{aligned}$$

Full-dimensionality and centering imply $A > 0$ and $B > 0$. Set $n = d+1$ and $N = n(n+1)/2$. For an ordered d -tuple $\mathbf{x}$, define

$$\begin{aligned}
F_{\mathbf{x}}(y) &= \det((x_1, 1), \dots, (x_d, 1), (y, 1)), \\
P(\mathbf{x}) &= \mathbb{E}(F_{\mathbf{x}}(X))_+, & N_-(\mathbf{x}) &= \mathbb{E}(-F_{\mathbf{x}}(X))_+.
\end{aligned}$$

Centering gives $D = 2\mathbb{E} \min(P, N_-) \geq 0$. For independent test points Y, Z define the nonnegative witness

$$\begin{aligned}
\psi_{\mathbf{x}}(Y, Z) &= \min\{(F_{\mathbf{x}}(Y))_+, (-F_{\mathbf{x}}(Z))_+\} \\
&\quad + \min\{(-F_{\mathbf{x}}(Y))_+, (F_{\mathbf{x}}(Z))_+\}.
\end{aligned}$$

Its expectation over all $d+2$ independent samples is at most D .

For anchors $W = (w_0, \dots, w_d)$, let $V(W)$ be the absolute lifted determinant, *not* the simplex volume $V(W)/d!$. Define $H(W)$ by integrating in x the sum of these N witnesses:

1. For every i , the base is the d other anchors and the tests are w_i, x .
2. For every $i < j$, the base is x and the $d-1$ anchors other than i, j , and the tests are w_i, w_j .

Order bases by a fixed convention; ψ is unaffected by changing the overall determinant sign. Each summand uses $d+2$ independent samples before conditioning, so $\mathbb{E}H \leq ND$.

Selection without an expectation-of-ratio error

On anchor tuples define the probability measure

$$d\mathbb{P}_V(W) = \frac{V(W)}{B} d\nu^{\otimes(d+1)}(W).$$

This is a probability because $\mathbb{E}V = B$. It gives full mass to support tuples with $V > 0$. Define H/V only there. Then

$$\mathbb{E}_{\mathbb{P}_V}(H/V) = \frac{\mathbb{E}[H\mathbf{1}_{\{V>0\}}]}{B} \leq \frac{ND}{B}.$$

Consequently some support tuple with $V > 0$ satisfies $H/V \leq ND/B$. Indeed, if the integrable ratio were strictly larger everywhere on a full-measure set, its expectation would be strictly larger too. No compact minimum on the open set $\{V > 0\}$ is assumed. When $D = 0$, the ratio is zero $\mathbb{P}_V$ -almost surely.

Pointwise clipped-witness bound

Fix such a tuple and let $\alpha_i(x)$ be its affine barycentric coordinates. Let $r(x)$ be the first index attaining the largest positive coefficient; some positive coefficient exists because $\sum_i \alpha_i = 1$. This is Borel measurable. Put

$$\varphi(x) = \sum_i \min\{1, (-\alpha_i(x))_+\} + \sum_{i < j} \min\{(\alpha_i(x))_+, (\alpha_j(x))_+\}.$$

The first witnesses equal $V \min\{1, (-\alpha_i)_+\}$. When $\alpha_i, \alpha_j > 0$, the pair witness is $V \min(\alpha_i, \alpha_j)$, because the two tested determinants have opposite signs. When either is nonpositive, the displayed lower bound is zero. Thus $V\mathbb{E}\varphi \leq H$.

Let $\Delta = \text{diam}(\text{supp } \nu)$ in this subsection. If some $\alpha_i \leq -1$, then $\varphi \geq 1$ and $\|x - w_r\| \leq \Delta \leq \Delta\varphi$. Otherwise every negative magnitude is below 1, so clipping is inactive. Let

$$N_x = \sum_i (-\alpha_i)_+, \quad U_x = \sum_{i \neq r} (\alpha_i)_+.$$

The pairs containing r give $U_x \leq \sum_{i < j} \min\{(\alpha_i)_+, (\alpha_j)_+\}$. Also

$$x - w_r = \sum_{i \neq r} \alpha_i(x)(w_i - w_r),$$

so $\|x - w_r\| \leq \Delta(N_x + U_x) \leq \Delta\varphi$. Therefore, in all cases,

$$h := \mathbb{E}\|X - w_{r(X)}\| \leq \Delta\mathbb{E}\varphi \leq \Delta H/V \leq \Delta ND/B.$$

This is the entire stronger assignment theorem. In the Euclidean unit ball, $\Delta \leq 2$, giving $h \leq (d+1)(d+2)D/B$.

Quantitative origin inclusion when dispersion is supplied

If $\mathbb{E}(u \cdot X)_+ \geq \eta$ for every unit u and $h \leq \eta - b$ with $b > 0$, then the assigned law has $\mathbb{E}(u \cdot w_r)_+ \geq b$. At least one anchor has $u \cdot w_i \geq b$ in every direction. Thus $bB_2^d \subset T = \text{conv}(w_i)$. This implication is separate from the assignment theorem and is exactly where cone-body geometry will enter.

2 Prescribed-simplex normalization and two elementary improvements

Fix the originally prescribed maximum simplex S ; do not select a replacement maximum. Apply an invertible affine map sending S to the regular simplex Δ centered at zero with inradius 1. Its vertices v_i have $\|v_i\| = d$ and $v_i \cdot v_j = -d$ for $i \neq j$.

For every $x \in K$, replacing vertex i by x changes the simplex volume by the factor $|\alpha_i(x)|$. Maximality gives $|\alpha_i(x)| \leq 1$. Since $\sum_i \alpha_i = 1$,

$$\begin{aligned} \|x\|^2 &= d \left[(d+1) \sum_i \alpha_i(x)^2 - 1 \right] \\ &\leq d[(d+1)^2 - 1] = d^2(d+2). \end{aligned}$$

Therefore

$$B_2^d \subset \Delta \subset K \subset RB_2^d, \quad R = d\sqrt{d+2}.$$

Also, dilation of Δ by $1+t$ about its centroid is equivalent to $\alpha_i \geq -t/(d+1)$. The same $|\alpha_i| \leq 1$ immediately yields the global bound

$$E(K, \Delta) \leq d+1.$$

This is stronger than using $K \subset R\Delta$ for the global branch.

3 Actual cone-law dispersion without arbitrary-law substitution

For this normalized convex body, define ν as the pushforward of $h_K(u)dS_K(u)/(d|K|)$ under $u \mapsto u/h_K(u)$. The classical identities $\int h_K dS_K = d|K|$ and $\int u dS_K = 0$ make it a centered probability law, supported on $\partial K^\circ \subset B_2^d$. Its support spans $\mathbb{R}^d$ because every orthogonal projection of K has positive $(d-1)$ -volume. The source cone representation [2, Proposition 4.1] gives

$$a(K) = \frac{B}{(d+1)A}, \quad \frac{D}{B} = \frac{(d+1)e}{1+(d+1)e}.$$

Thus Section 1 gives anchors and a measurable assignment satisfying

$$h \leq \frac{Ce}{1+(d+1)e} \leq Ce, \quad C = (d+1)^2(d+2).$$

Cauchy's projection formula, including its factor 1/2, gives

$$\mathbb{E}|u \cdot X| = \frac{2\pi_K(u)}{d|K|}, \quad \mathbb{E}(u \cdot X)_+ = \frac{\pi_K(u)}{d|K|}$$

for unit u , where $\pi_K(u) = |K \mid u^\perp|_{d-1}$ and centering gives the second equality.

The geometric lower bound is particularly simple. Slice K by lines parallel to u over $K \mid u^\perp$. Every nonempty chord has length at most the directional width $w_K(u) = h_K(u) + h_K(-u)$. Fubini therefore gives

$$|K| \leq w_K(u)\pi_K(u) \leq 2R\pi_K(u).$$

It follows that

$$\mathbb{E}(u \cdot X)_+ \geq \frac{1}{2dR} = \frac{2}{M}, \quad M = 4dR.$$

This is valid for actual convex-body cone laws. It is not asserted for arbitrary centered laws merely because their support lies in a ball.

If $h \leq 1/M$, Section 1 implies $M^{-1}B_2^d \subset T \subset B_2^d$. Hence $P = T^\circ$ is a genuine enclosing simplex with

$$B_2^d \subset K \subset P \subset MB_2^d.$$

Each selected anchor lies on ∂K° , so $h_K(w_i) = h_P(w_i) = 1$.

4 Complete assignment-to-projection conversion

Let q_i be the vertex of P opposite the facet with polar normal w_i , and let $\lambda_i > 0$ be the barycentric coordinates of zero in T . The cone law of P is $\sum_i \lambda_i \delta_{w_i}$: it is supported on exactly these facet polar images, is centered, and has unique barycentric weights.

Let $p_i = \mathbb{P}(r(X) = i)$ and $c = \sum_i p_i w_i$. Centering gives $\|c\| \leq h$. The barycentric functions of T are

$$\alpha_i(x) = \lambda_i(1 - q_i \cdot x).$$

Thus $p_i - \lambda_i = -\lambda_i q_i \cdot c$ and $\sum_i |p_i - \lambda_i| \leq Mh$. Comparing the assigned law and ν_P , for every unit u ,

$$|\mathbb{E}_\nu|u \cdot X| - \mathbb{E}_{\nu_P}|u \cdot X|| \leq (1 + M)h,$$

and consequently

$$\left| \frac{\pi_K(u)}{|K|} - \frac{\pi_P(u)}{|P|} \right| \leq \frac{d}{2}(1 + M)h.$$

Since h_P is M -Lipschitz and $h_P(w_i) = 1$,

$$0 \leq z := \int (h_P(x) - 1) d\nu(x) \leq Mh \leq 1.$$

By homogeneity in $x = u/h_K(u)$, the integral equals $V(K[d-1], P)/|K| - 1$. Minkowski's first inequality, equivalently the first variation of Brunn–Minkowski, yields

$$\frac{|P|}{|K|} \leq (1 + z)^d, \quad |P| - |K| \leq d2^{d-1}Mh|K|.$$

Because $\text{supp } \nu_P \subset B_2^d$, we have $\pi_P(u)/|P| \leq d/2$. Combining these facts and $|K| \leq (2R)^d$ proves

$$0 \leq \pi_P(u) - \pi_K(u) \leq Jh$$

with J as stated. Nonnegativity uses the genuine inclusion $K \subset P$.

5 An improved projection cap using the radius of K

Suppose $B_2^d \subset K \subset P$, $K \subset RB_2^d$, and all projection deficits are at most ε . Let $s = d_H(K, P)$. The conclusion is

$$s \leq 2\sqrt{d-1}(R+1)\varepsilon^{1/(d-1)}$$

whenever $\sqrt{d-1}\varepsilon^{1/(d-1)} \leq 1/2$. No bound on the outer radius of P is needed for this statement, apart from compactness.

For $s > 0$, choose a farthest $q \in P$, its nearest $k \in K$, and $n = (q - k)/s$. The projection optimality condition gives $n \cdot k = h_K(n)$. Crucially,

$$\|q\| \leq R + s.$$

Choose a unit $u \perp n$ and project onto $u^\perp$, of dimension $m = d - 1$. Let $\tau = s/[2(R + s + 1)] < 1$. The projected P contains the ball $(1 - \tau)\bar{q} + \tau B_2^m$. Its lowest n -coordinate is at least

$$n \cdot q - \tau(\|q\| + 1) \geq h_K(n) + s/2,$$

so the ball is disjoint from projected K . It contains a cube of side $2\tau/\sqrt{m}$, hence

$$\varepsilon \geq \left[\frac{s}{\sqrt{m}(R + s + 1)} \right]^m.$$

Putting $t = \sqrt{m}\varepsilon^{1/m}$, we have $s \leq t(R + s + 1)$. If $t \leq 1/2$, this rearranges to $s \leq 2t(R + 1)$. The case $\varepsilon = 0$ follows from the same cap inequality. This proves the claim.

This step avoids transporting the large polar radius M into every geometric factor. Hereafter $L = 2\sqrt{m}(R + 1)$.

6 Local gate and retention of the original maximum simplex

Assume $0 \leq e \leq e_0$. Since $J \geq M$ and $8dL \geq 1$,

$$h \leq Ce \leq \frac{1}{J(8dL)^m} \leq \frac{1}{M}.$$

Section 4 applies with $\varepsilon = Jh$. Write $y = (JCe)^{1/m}$. Then

$$y \leq \frac{1}{8dL}, \quad \sqrt{m}y \leq \frac{1}{16d(R + 1)} \leq \frac{1}{2}.$$

The improved cap gives

$$s \leq L(Jh)^{1/m} \leq Ly \leq \frac{1}{8d}.$$

In particular $P \subset (R + s)B_2^d \subset (R + 1)B_2^d$. Also $h_K \geq h_P - s \geq (1 - s)h_P$, because $h_P \geq 1$, so $(1 - s)P \subset K$.

The fixed original Δ is maximum. Therefore

$$\frac{|\Delta|}{|P|} \geq (1 - s)^d \geq 1 - ds.$$

Put $\delta = ds \leq 1/8$. Express the vertices of Δ in barycentric coordinates in P , producing a nonnegative column-stochastic $(d + 1) \times (d + 1)$ matrix W with $|\det W| = |\Delta|/|P| \geq 1 - \delta$. Hadamard gives each column norm at least $1 - \delta$, and its largest coordinate is at least the squared norm, hence at least $1 - 2\delta$.

These dominant coordinates occupy distinct rows: if two occupied the same row, the independent-row-selection interpretation of the permanent would give

$$|\det W| \leq \text{per}(W) \leq 1 - (1 - 2\delta)^2 \leq 4\delta < 1 - \delta,$$

a contradiction. Match vertices using these distinct rows. Each vertex of Δ differs from its matched vertex of P by at most

$$2\delta \operatorname{diam}(P) \leq 4(R+1)ds.$$

Every vertex of P is matched. Thus

$$K \subset P \subset \Delta + 4(R+1)ds B_2^d \subset [1 + 4(R+1)ds]\Delta,$$

using $B_2^d \subset \Delta$. This proves

$$E(K, \Delta) \leq 4(R+1)dL(JC)^{1/m}e^{1/m} = \frac{G_d}{2}e^{1/m}.$$

All inequalities include $e = 0$: then $h = 0$, projection deficits vanish, $s = 0$, and the same matching identifies the simplices.

7 Global branch and original centroid

By the exact definitions,

$$G_d e_0^{1/m} = R + 1 \geq d + 1.$$

For $e > e_0$, the universal maximality bound $E(K, \Delta) \leq d + 1$ therefore gives $E(K, \Delta) \leq G_d e^{1/m}$. For $e \leq e_0$, Section 6 applies. This closes the global theorem without any enormous minimum threshold entering the final constant.

The affine normalization used the prescribed S throughout. An invertible affine map sends its centroid to the centroid of Δ and intertwines homotheties about these centroids. Pulling back the final containment gives exactly the stated dilation about z_S . The source invariant a , hence e , is affine invariant [2, 3]. No assertion that an original Euclidean unit ball is preserved under affine pullback is needed.

8 Explicit polynomial bound for the coefficient

For $d \geq 3$,

$$\begin{aligned} C &\leq 3d^3, & M &\geq 1, & 1 + M + d2^{d-1}M &\leq d2^d M, \\ J &\leq d^2 M(4R)^d, & R + 1 &\leq 2R, & L &\leq 4\sqrt{d} R. \end{aligned}$$

Consequently

$$\begin{aligned} G_d &\leq 256d^{3/2} R^3 [12d^5 M R]^{1/(d-1)} \\ &= 256d^{3/2} R^3 [48d^6 R^2]^{1/(d-1)}. \end{aligned}$$

Since $R^2 = d^2(d+2) \leq (5/3)d^3$,

$$G_d \leq 256(5/3)^{3/2} d^6 [80d^9]^{1/(d-1)}.$$

For integer $d \geq 3$, $d \leq 3^{(d-1)/2}$: this follows by induction, since $(d+1)/d \leq 4/3 < \sqrt{3}$. Therefore

$$[80d^9]^{1/(d-1)} \leq \sqrt{80} 3^{9/2} < 1280.$$

Also $(5/3)^{3/2} < 9/4$. Hence

$$G_d < 576 \cdot 1280 d^6 = 737280 d^6 < 2^{20} d^6.$$

This polynomial is deliberately nonoptimized. The exact displayed G_d is substantially smaller in many dimensions.

9 Sharpness and remaining scope

The exponent cannot improve: the pinned equal-intercept simplex truncations have

$$E = (d+1)t, \quad e \sim \frac{d(d-1)}{(d+1)^2} t^{d-1} \quad (t \downarrow 0)$$

for the maximum simplex used in the obstruction [1, 4]. The direct actual-polytope construction proved in Appendix A gives the stronger global coefficient obstruction

$$G_d \geq (d+1)[d(d+1)]^{1/(d-1)}.$$

Its order is $d+1+2\log d+O((\log d)^2/d)$, leaving a gap between linear and d^6 growth. It rules out a dimension-free coefficient, while the theorem above rules out the necessity of superpolynomial dimension dependence for this invariant and every-maximum-simplex metric.

The abstract assignment theorem controls mean transportation distance, not support Hausdorff distance. Rare atoms can remain far away while carrying negligible mass. The cone identities, centering, enclosing inclusion, first variation, and projection caps are essential to the geometric conversion. The companion `weighted_anchors.md` in the frozen release basis gives the detailed arbitrary-law theorem and exact stress tests.

The frozen refinement records an independent analytic model review. Such review is not human peer review or Lean formalization; no priority claim or result about a different projection-body invariant is made. The cone-law identity and affine invariance are imported from the pinned source; the classical Cauchy, surface-area, first-variation, and Brunn–Minkowski identities are explicitly identified inputs.

The release basis intentionally omits separate anisotropic and simultaneous-truncation research. No theorem from that material is needed here or endorsed by this release basis.

A focused next target is $G_d = O(d)$, which remains open here. The exact G_d above is asymptotic to $64d^6$. The remaining loss lies in Euclidean cap and retention geometry. A relative cap estimate intrinsic to the selected polar simplex could be investigated using the general-norm assignment theorem; such an improvement is not proved here.

10 Sources and literature positioning

The exact source bytes and public commit URLs are recorded in the accompanying source-pin file. The original sharp manuscript is [1]; entry005 v3 [2] and the released explicit modulus [3] supply the invariant’s cone representation, its affine invariance, and the older stability argument. This argument changes anchor selection and dimension estimates, not the definition of the invariant.

The frozen refinement records a primary-literature inspection on 7 October 2026 with the following scope:

- Böröczky’s *The stability of the Rogers–Shephard inequality and of some related inequalities* [6] has a Theorem 3 concerning $|K|^{d-1}|\Pi^*K|$, the polar projection-body volume, and a Banach–Mazur distance with a selectable simplex. Its stated upper modulus is $d^{88d}\varepsilon^{1/d}$. This is a different deficit from $e(K)$; no comparison identifying them has been proved here. Therefore the present refinement is not asserted to improve that theorem.
- Böröczky and Henk’s *Cone-volume measure and stability* [7] concerns cone-volume concentration and the U -functional, rather than the first absolute determinant difference used here. It does not by itself give a realization theorem for arbitrary centered laws on prescribed polar support points.

- Böröczky, Fodor and Hug’s *Strengthened inequalities for the mean width and the ℓ -norm* [8] studies different extremal functionals and John/Löwner positioning. It provides context, not an imported estimate for $e(K)$.

The targeted primary-source search recorded in the frozen refinement did not establish a prior theorem identical to the weighted first-moment assignment theorem or the polynomial constant above. This is not an exhaustive literature search and does not justify a novelty claim.

A A stronger realized lower bound for the endpoint coefficient

For every $d \geq 3$, a universal coefficient in the every-maximum-simplex estimate must satisfy

$$G_d \geq (d+1)[d(d+1)]^{1/(d-1)}.$$

This is realized by the explicit full-dimensional polytope and maximum simplex below. The frozen companion proof records an independent analytic audit of the construction, maximality, barycentric excess, invariant calculation, and direct facet certificate. No arbitrary law is treated as a convex-body realization. The generic facet formula is from the pinned entry005 v3, Section 1 [2]: if u_i is facet area times outward unit normal and b_i is facet area times support number, then

$$H = \sum_{|I|=d} |\det(u_i : i \in I)|, \quad L = \sum_{|I|=d+1} |\det((u_i, b_i) : i \in I)|,$$

$$a(K) = \frac{L}{d|K|H}.$$

Here H and L are facet-minor sums local to this appendix, not the assignment witness H or cap constant L of the main proof. The optional pyramid calculation provides a second route; the direct certificate avoids needing that recursion as an extra input.

A.1 An elementary global bound on the geometric excess

If $S = \text{conv}(v_0, \dots, v_d)$ is maximum-volume in K , its barycentric coordinates satisfy

$$|\alpha_i(x)| \leq 1 \quad (x \in K).$$

Indeed, replacing v_i by x multiplies volume by $|\alpha_i(x)|$. Centroid dilation has the exact characterization

$$x \in z_S + (1+t)(S - z_S) \iff \alpha_i(x) \geq -\frac{t}{d+1} \quad \text{for every } i.$$

Consequently

$$E(K, S) \leq d+1. \tag{1}$$

This applies to every maximum simplex without a normalization or a small-deficit hypothesis. It is much sharper than the coarse outer-radius bound used as the large-deficit branch in the supplied proof.

A.2 The square-pyramid construction

Take

$$C_d = \text{conv}(0, e_1, e_2, e_1 + e_2, e_3, \dots, e_d), \quad S_d = \text{conv}(0, e_1, \dots, e_d).$$

This is the standard unit square followed by $d - 2$ pyramids. The square has $a = 1/2$, and each pyramid adds one to $1/a$, so

$$a(C_d) = \frac{1}{d}, \quad e(C_d) = \frac{1}{d(d+1)}. \quad (2)$$

To check maximality directly, a full-dimensional vertex simplex must contain every $e_3, \dots, e_d$, since no other vertex has a nonzero corresponding coordinate. It must also contain exactly three square vertices. Every such triangle has area $1/2$, so every full-dimensional vertex simplex has volume $1/d!$. The absolute determinant is separately convex in its vertices, hence no simplex with arbitrary vertices in C_d has larger volume. In particular, S_d is maximum.

The point $e_1 + e_2$ has barycentric coordinates

$$(-1, 1, 1, 0, \dots, 0)$$

in S_d . Hence $E(C_d, S_d) \geq d + 1$, with equality by (1). Combining these identities proves the stated necessary lower bound for every universal coefficient.

There is also a direct facet certificate for (2), independent of the pyramid recursion. Put $c = 1/(d-1)!$. The facet area-normal/support pairs are

$$\begin{aligned} &(-ce_1, 0), \quad (-ce_2, 0), \quad (-2ce_i, 0) \quad (3 \leq i \leq d), \\ &\left(c \left(e_1 + \sum_{i=3}^d e_i\right), c\right), \quad \left(c \left(e_2 + \sum_{i=3}^d e_i\right), c\right). \end{aligned}$$

The corresponding halfspaces describe the square cross-sections

$$0 \leq x_1, x_2 \leq 1 - \sum_{i=3}^d x_i.$$

Its volume is $2/d!$. Direct maximal-minor sums give

$$H = c^d d 2^{d-1}, \quad L = c^{d+1} 2^d,$$

so $a = L/(d|C_d|H) = 1/d$. To count L , only the four subsets obtained by omitting either top facet or either of the first two coordinate facets contribute, each by $c^{d+1} 2^{d-2}$. For H , the contributions with zero, one, or two top facets are respectively $1, d, d-1$, all multiplied by $c^d 2^{d-2}$.

The lower bound from the supplied small-truncation family is

$$L_d^{\text{small}} = (d+1) \left[\frac{(d+1)^2}{d(d-1)} \right]^{1/(d-1)}.$$

The square-pyramid bound is strictly larger since

$$\left(\frac{L_d^{\text{square}}}{L_d^{\text{small}}} \right)^{d-1} = \frac{d^2(d-1)}{d+1} > 1 \quad (d \geq 3).$$

For example, its values for $d = 3, 4, 5$ are approximately 13.8564, 13.5721, and 14.0421, respectively. Numerical values are illustrative only; the formula is exact.

References

- [1] *Sharp lower-end simplex stability from integrated witnesses and projection caps*. AI-assisted research supplement to entry005, 7 October 2026. Original sharp manuscript, pinned source: <https://github.com/mxym/math/blob/e4352dd1a3415ee0dbe3213e13932173d879fe10/notes/sharp-simplex-stability/proof.tex>.
- [2] mxym, *Random-determinant rigidity, sharp symmetric cone bounds, and spectral nonattainment*. AI-assisted research manuscript, entry005, version 3, 7 October 2026. Sections 1 and 4 give the invariant definition, facet formula, and cone-law representation. Pinned source: <https://github.com/mxym/math/blob/31e3d8a37e4a3051a7f5a2535be1c75642e15bcc/preprints/005-simplex-product-optimum/v3/paper.md>.
- [3] mxym, *Effective simplex rigidity for the projection cone invariant*. Supplement to entry005, 7 October 2026. Pinned source: <https://github.com/mxym/math/blob/87868bcb65bf0460a90d6bc2efc3324e46481230/notes/quantitative-projection-simplex-stability/paper.md>.
- [4] *An exact exponent obstruction for lower-end projection-cone simplex stability*. Research supplement to entry005; AI-assisted research note, 7 October 2026. Pinned source: <https://github.com/mxym/math/blob/87868bcb65bf0460a90d6bc2efc3324e46481230/notes/simplex-truncation-stability/proof.tex>.
- [5] R. Schneider, *Convex Bodies: The Brunn–Minkowski Theory*, second expanded edition, Cambridge University Press, 2014. Classical Cauchy, surface-area, first-variation, and Brunn–Minkowski inputs; reference inherited from the pinned source manuscripts.
- [6] K. Böröczky, Jr., *The stability of the Rogers–Shephard inequality and of some related inequalities*. Author-hosted manuscript: <https://www.renyi.hu/~carlos/rogerstab.pdf>.
- [7] K. J. Böröczky and M. Henk, *Cone-volume measure and stability*. Author-hosted manuscript: <https://www.renyi.hu/~carlos/cone-volume-stability.pdf>.
- [8] K. J. Böröczky, F. Fodor and D. Hug, *Strengthened inequalities for the mean width and the ℓ -norm*. <https://arxiv.org/abs/2001.10706>.